Kshitiz Kumar

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Summary

AI Engineer specializing in LLM systems, RAG pipelines, and inference optimization. Built and deployed production-grade AI applications from scratch. Experienced in taking AI systems from research to deployment.

Education

University Of Petroleum and Energy Studies (UPES), Dehradun **2026**
B.Tech, Big Data *Dehradun*
- **Coursework:** Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, Software Engineering, AI & ML

Work Experience

Cidroy Infotech **Dec 2025 - Present**
SDE-1 *Remote*
- Built an AI-driven automated quiz generation system using n8n and large language models.
- Optimized the assessment generation pipeline to reduce per-candidate test cost by ~99.7%, lowering it from ~$1 to ~$0.003 per test.
- Fine-tuned YOLOv8 model achieving 96% accuracy for OCR pipeline on custom dataset, enabling automated document processing
- Researched Vision Language Models (VLMs) for document understanding, evaluating modern approaches vs traditional OCR
- Investigated inference optimization for low-VRAM environments, quantization methods including GGUF, GPTQ, and bitsandbytes

Xebia **Jun 2025 - Jul 2025**
Project Intern *Remote*
- Developed interactive dashboards using Streamlit and Plotly to visualize patient vitals and real-time alerts.
- Configured secure and scalable infrastructure on AWS using IAM, S3, and EC2, enabling reliable platform deployment.
- Utilized Docker to containerize backend services, improving deployment consistency and supporting infrastructure observability using CloudTrail and centralized logging.

Projects

Retrieval-Augmented Generation (RAG) System from Scratch | Live Demo **Feb 2026**
- Built a Retrieval-Augmented Generation (RAG) pipeline from scratch without frameworks, implementing custom chunking, embedding, similarity scoring, and retrieval logic.
- Implemented cosine similarity–based retrieval with explicit top-k and similarity threshold gating to prevent hallucinations.
- Integrated local Hugging Face Seq2Seq models (FLAN-T5) for deterministic, grounded inference with controlled generation settings.
- Implemented programmatic refusal logic to block LLM generation when relevant context is missing.
- Tech Stack: Python, Hugging Face Transformers, PyTorch, FAISS, Groq
- Source - https://github.com/NotKshitiz/rag-from-scratch

GPU-Aware ML Inference Engine **Jul 2025 - Aug 2025**
- Developed a plug-and-play Python Desktop application that auto-converts Hugging Face causal language models (llama-cpp compatible) to GGUF format for local inference.
- Enabled VRAM (NVIDIA) GPU-based inference using Pytorch, with fallback to CPU when GPU is unavailable.
- Built support for seamless prompt-based text generation via a simple FastAPI UI and tokenizer auto-detection.
- Tech Stack: Python, Pytorch, FastAPI, Hugging Face, Llama-cpp, Electron
- Source: https://github.com/NotKshitiz/paradigmai

k8scan - Kubernetes Observability & Debugging CLI Tool **Jun 2025**
- Built a CLI tool in Golang (Cobra) to identify and manage Kubernetes issues like CrashLoopBackOff, orphaned services, and unstable pods.
- Benchmarked on 60+ pods (14 crashlooping) detecting 25% more unstable pods through deeper status parsing than kubectl
- Integrated with Prometheus, Grafana, Loki, and Alertmanager for metrics, logs, and alerting.
- Tech Stack: Golang, Cobra CLI, Kubernetes, Prometheus, Grafana, Loki, Alertmanager
- Source: https://github.com/NotKshitiz/k8scan

Skills

- **Languages & Scripting**: Python, C++/CUDA, Java, Bash
- **Machine Learning & Deep Learning**: Pytorch, Transformers, Hugging Face, Large Language Models, RAG, LangChain, ChromaDB, FAISS, RAGAS, GGUF
- **Automation & Workflow Orchestration**: n8n
- **Cloud & DevOps**: AWS, Docker, Kubernetes, Jenkins
- **Monitoring & Observability**: Prometheus, Grafana, Loki
- **Mathematical Foundations**: Linear Algebra, Probability & Statistics, Information Theory, Numerical Computation